

Aalok Gangopadhyay

SCIENTIST · TATA CONSULTANCY SERVICES RESEARCH

☎ (+91) 9558210714 | ✉ aalok1993@gmail.com | 🏠 aalok1993.github.io | 📱 aalok1993 | 🌐 aalok-gangopadhyay

Professional Summary

Aalok Gangopadhyay is a Scientist at TCS Research in the Visual Computing and Embodied AI group, where he leads a team working on two problems: (1) 3D Generative AI - interactive generation of 3D shapes through an intuitive extended reality interface driven by multimodal inputs such as hand-gesture sketches and voice prompts; and (2) static and dynamic scene generation from one or a few images for interactive extended reality experiences.

Previously, he was a Visiting Fellow at the Tata Institute of Fundamental Research (TIFR) under Prof. Hariharan Narayanan, working on Manifold Learning and Algebraic Combinatorics. This work included a manifold denoising algorithm with theoretical guarantees, along with results on combinatorial hives — estimating limiting surfaces of GUE hives, lozenge tilings, and surface tension.

He earned his Ph.D. and B.Tech. from the Indian Institute of Technology Gandhinagar (IITGN), advised by Prof. Shanmuganathan Raman, specializing in Computer Graphics, Computer Vision, Deep Learning, and Shape Analysis. There he designed invertible neural networks and neural implicit representations for parameterizing homeomorphisms and diffeomorphisms, addressing problems such as Escherization with isohedral tilings, density estimation on identification spaces, optimal embedding of knots for shadow art, and knot untangling. He also developed differentiable rendering algorithms for tubular knots and hand models for shadow art, a graph neural network approach to mesh denoising, correspondence, and temporal blending, and achieved state-of-the-art results in motion deblurring and dynamic scene classification. His work has appeared in venues including CGF, SGP, ICCV, Eurographics, ICPR, GlobalSIP, NCC and AIHPC.

Research Interests: Computer Graphics, Computer Vision, Deep Learning, Shape Analysis, Multimodal AI, Inverse Rendering

Employment

Tata Consultancy Services Research

Scientist, Visual Computing and Embodied AI

- 3D Generative AI and Scene Reconstruction

Kolkata, India

Oct 2025 - Present

Tata Institute of Fundamental Research

Visiting Fellow, School of Technology and Computer Science

- Advisor: Dr. Hariharan Narayanan

Mumbai, India

Sept 2023 - Sept 2025

Education

Indian Institute of Technology Gandhinagar

PhD, Electrical Engineering

- Advisor: Dr. Shanmuganathan Raman
- Thesis: Deep Learning based Representation, Processing and Inverse Rendering of Curves and Surfaces

Gandhinagar, India

May 2017 - June 2023

Indian Institute of Technology Gandhinagar

B.Tech, Electrical Engineering with minors in Computer Science

- CGPA 9.17/10

Gandhinagar, India

July 2011 - May 2015

Awards & Achievements

- Received best poster presentation award in Pacific Graphics 2023.
- Awarded TCS Research Fellowship (July 2018 - June 2022).
- Qualified for Indian National Physics Olympiad (INPhO) 2011 after clearing National Standard Examination in Physics (NSEP).
- Secured All India Rank (AIR) 2812 in IIT-JEE Advanced 2011.

Technical Skills

- Programming Languages: C, C++, Python, Matlab
- Libraries and Platforms: Tensorflow, PyTorch, OpenCV, Blender
- Operating systems: Linux, MacOSX, Windows

Publications

ACCEPTED

- **Gangopadhyay, Aalok**, and Hariharan Narayanan. "On the randomized Horn problem and the surface tension of hives." Ann. Inst. Henri Poincaré Comb. Phys. Interact. (2025)
- **Gangopadhyay, Aalok**, Paras Gupta, Tarun Sharma, Prajwal Singh, and Shanmuganathan Raman. "Search Me Knot, Render Me Knot: Embedding Search and Differentiable Rendering of Knots in 3D." In COMPUTER GRAPHICS forum, vol. 43, no. 5. 2024. Presented in Symposium on Geometry Processing (SGP), 2024.
- **Gangopadhyay, Aalok**, Prajwal Singh, Ashish Tiwari, and Shanmuganathan Raman. "Hand Shadow Art: A Differentiable Rendering Perspective." The Eurographics Association, 2023. Presented in Pacific Graphics 2023.
- **Gangopadhyay, Aalok**, Prajwal Singh, and Shanmuganathan Raman. "APEX-net: automatic plot extraction network." 2022 National Conference on Communications (NCC). IEEE, 2022.
- **Gangopadhyay, Aalok**, Shashikant Verma, and Shanmuganathan Raman. "DMD-Net: Deep Mesh Denoising Network." 2022 26th International Conference on Pattern Recognition (ICPR). IEEE, 2022.
- Ramakrishnan, Sainandan, Shubham Pachori, **Aalok Gangopadhyay**, and Shanmuganathan Raman. "Deep generative filter for motion deblurring." In Proceedings of the IEEE International Conference on Computer Vision Workshops, pp. 2993-3000. 2017
- **Gangopadhyay, Aalok**, Shubham Pachori, and Shanmuganathan Raman. "Automatic silhouette photography." In 2016 Twenty Second National Conference on Communication (NCC), pp. 1-6. IEEE, 2016.
- **Gangopadhyay, Aalok**, Shivam Mani Tripathi, Ishan Jindal, and Shanmuganathan Raman. "Dynamic scene classification using convolutional neural networks." In 2016 IEEE Global Conference on Signal and Information Processing (GlobalSIP), pp. 1255-1259. IEEE, 2016.

PREPRINTS

- Fefferman, Charles, **Aalok Gangopadhyay**, Matti Lassas, Jonathan Marty, and Hariharan Narayanan. "Denoising data using convex relaxations." arXiv preprint arXiv:2605.02327 (2026).
- Bhardwaj, Satyam, Lokender Tiwari, Shivam Shukla, Mritunjoy Halder, **Aalok Gangopadhyay**, and Brojeshwar Bhowmick. "UltraNormal: Scaling Pretrained Diffusion-based Surface Normal Estimators to Ultra-High Resolution."
- **Gangopadhyay, Aalok**, Dwip Dalal, Progyan Das, and Shanmuganathan Raman. "Flow Symmetrization for Parameterized Constrained Diffeomorphisms." arXiv preprint arXiv:2312.06317 (2023).
- **Gangopadhyay, Aalok**, Abhinav Narayan Harish, Prajwal Singh, and Shanmuganathan Raman. "A Graph Neural Network Approach for Temporal Mesh Blending and Correspondence." arXiv preprint arXiv:2306.13452 (2023).

Internships and Research Positions

Deep Learning algorithms for detection and recognition of food items inside a refrigerator

Innit Inc. and IIT Gandhinagar

SUPERVISOR : DR. SHANMUGANATHAN RAMAN (IIT GANDHINAGAR), DR. HRISTO BOJINOV (CTO, INNIT INC.)

June 2016 - May 2017

- A deep learning algorithm was designed that performed object detection to extract the information about the food objects inside a refrigerator environment.
- A sticker based visual encoding-decoding scheme was developed that allowed us to faithfully encode information onto a physical sticker and decode that information content when viewed by a camera.

Junior Research Fellow in the field of Computer Vision and Deep Learning

IIT Gandhinagar

SUPERVISOR : DR. SHANMUGANATHAN RAMAN

July 2015 - June 2016

- Automatic Silhouette Photography: An algorithm for detecting the silhouette in a photograph was developed. This was complemented by a color transform mechanism that made the image aesthetically appealing.
- LDR to HDR: A dataset of HDR images was curated from various online sources. These HDR images were converted to LDR images with varying exposure time. A deep learning network was trained to map LDR images to HDR images.
- Image Denoising using Sparse Coding and non-local means: A large scale patch database was generated using images from the ImageNet and Places dataset. WTA (Winner Takes All) hash was used for computationally efficient implementation of a similarity based patch retrieval.
- Image morphing using Autoencoders with representational constraints in the latent space: The representational constraint models the relation between three images in the latent space while the decoder simultaneously tries to generate the images from the latent features.

Discovering rare transient event using data catalogs from CRTS sky survey

Caltech

SUPERVISOR : DR. ASHISH MAHABAL

May 2014 - July 2014

- We develop filtering algorithms to search for rare transient events by analyzing billions of uncategorized transient events present in Catalina Real-Time Transient Survey (CRTS). This also involved cross comparison with other sky survey catalogs. The list of potential events was reduced from billions to an order of thousands. An annotation tool was designed for citizen science, where potential events obtained after the filtration process could be labelled as a potential rare transient event upon visual inspection.

Development of Android applications

Grid Ants Pvt. Ltd.

SUPERVISOR : SARTHAK JAIN, PRATHAMESH JUVATKAR, PRASHANT BORDE

May 2013 - July 2013

- Development of front end (UI + UX) and the backend (database integration) of android applications for various clients.
- Designing of an audio recommendation engine, with server based playback methodology.

- A stochastic method was implemented for plotting various fractals using Iterated Function Systems. A program was designed in C/CUDA that generated Mandelbrot sets and Julia sets using Escape time algorithm. The program included added functionality of zooming, panning and varying the color patterns during run-time.
- Parallel computing algorithms for a variety of numerical methods were implemented using MPI, OpenMP, Pthreads, and CUDA on multicore systems. This was followed by a detailed performance analysis of the algorithms in terms of speedup, efficiency, etc. The performance metrics of the HPC Machine (SMA Hydra Linux Cluster) was estimated in terms of bandwidth, message latency and startup cost.

Teaching Experience

Assisted in teaching the following courses at IIT Gandhinagar:

- Deep Learning, (Spring 2022, Spring 2021)
- Probability and Random Processes, (Fall 2021, Fall 2019, Fall 2018)
- 3D Computer Vision, (Spring 2020)
- Signals and Systems, (Spring 2018)
- Mathematical Foundation for Computer Vision and Computer Graphics, (Fall 2017)

Invited Talks and Seminars

- Experiments with Random Hives, TIFR, August 2024
- Differentiable Algorithms for Representation, Processing and Rendering of Shapes, IIT Bombay, August 2023
- Differentiable Algorithms for Representation, Processing and Rendering of Shapes, TIFR, June 2023
- Mesh Denoising using Graph Neural Networks, TCS Virtual Research Cafe, August 2022
- GANs for Computer Vision, NASSCOM CoE xChange Program, IIT Gandhinagar, November 2019
- Introduction to Deep Learning, CSE Seminar, IIT Gandhinagar, February 2017

Professional Services

- Organizer, Workshop on "AI in Spatial Computing: AR/VR, Metaverse", co-located with 11th International Conference on Pattern Recognition and Machine Intelligence (PReMI), IIT Delhi, December 2025.
- Reviewer, Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP)
- Reviewer, International Conference on Computer Vision and Image Processing (ICCVIP)

References

Dr. Brojeshwar Bhowmick

Principal Scientist and Head of Visual Computing and Embodied AI Research Area
TCS Research, Kolkata, India

Dr. Hariharan Narayanan

Associate Professor, School of Technology and Computer Science
Tata Institute of Fundamental Research, India

Dr. Shanmuganathan Raman

Professor and Head of the Department of Computer Science and Engineering, Professor in Electrical Engineering
Indian Institute of Technology, Gandhinagar, India